

Implement Client Script & UI Policy (Incident)

Team ID : SWTID-2026-1991

Team Size : 5

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Problem Statement:

Incident records require consistent and accurate data entry to ensure effective triage, routing, and resolution. Relying solely on user awareness and manual checks can lead to incomplete, inconsistent, or incorrect data being submitted, which affects reporting accuracy, SLA compliance, and overall service quality. There is a need to enforce conditional field behaviour and validation directly at the user interface level, using standardised UI Policies and Client Scripts to control field visibility, mandatory status, and save-time validation.

User Story:

As a service desk administrator, I want Incident forms to enforce field rules automatically so that high impact incidents are always created with the required details, such as an assignment group, an assigned user, and the correct urgency. This will ensure that Incidents meet defined business rules before they are submitted, reducing incomplete records and manual follow-up.

Project Objective:

The objective of this project is to demonstrate how ServiceNow client-side controls can be used to enforce data integrity on Incident records. By using UI Policies and Client Scripts, the project aims to:

1. Dynamically make fields mandatory or read-only based on the Impact of the Incident.
2. Auto-populate values, such as setting Urgency to High when Impact is High.
3. Prevent record submission when required conditions are not met.
4. Block direct list edits of the State field to maintain data consistency.

Skills:

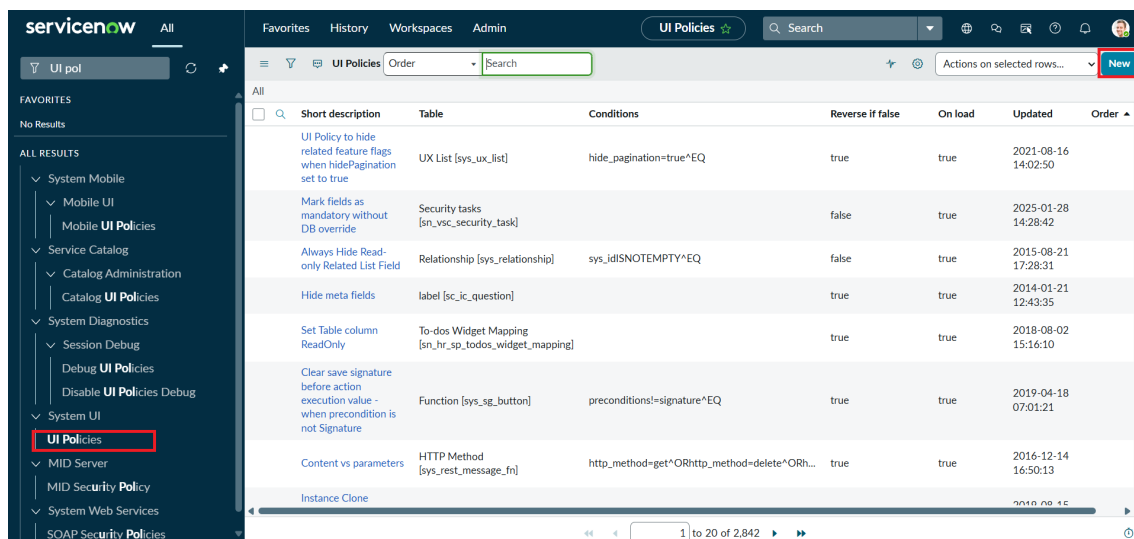
- Incident Management
- UI Policy
- UI Policy Actions
- Client Scripts
- Form Validation

TASK INITIATION

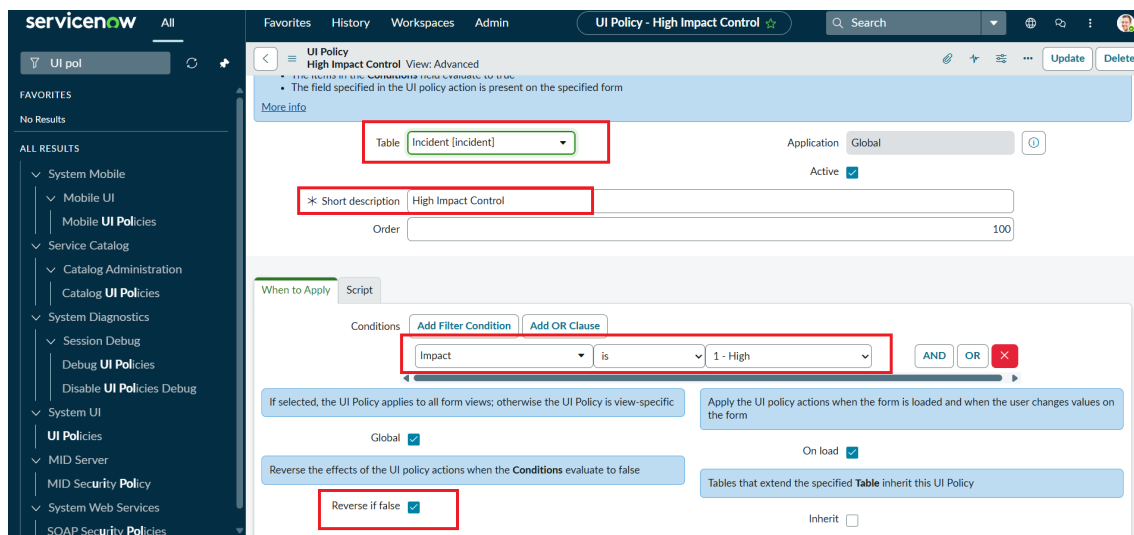
Milestone 1 : UI Policy

Activity 1: Create UI Policy on Incident (High Impact Control)

- Log in to the ServiceNow instance with administrative or configuration access.
- Click on All >> search for System UI >> click on UI Policies.
- Click on New to create a new UI Policy record.
- Fill in the Short description as “High Impact Control”.
- Set Table as “Incident [incident]”.
- Keep Active as true so the policy is enforced on the Incident form.
- Under When to Apply >> Conditions, select Field as “Impact”, Operator as “is”, and Value as “1 - High”.
- Check the Reverse if false option, so the policy actions are undone when the condition is not met.
- Review the configuration and click on Submit to save the UI Policy.



Short description	Table	Conditions	Reverse if false	On load	Updated	Order
UI Policy to hide related feature flags when hidePagination set to true	UX List [sys_ux_list]	hide_pagination=true*EQ	true	true	2021-08-16 14:02:50	
Mark fields as mandatory without DB override	Security tasks [sn_vsc_security_task]		false	true	2025-01-28 14:28:42	
Always Hide Read-only Related List Field	Relationship [sys_relationship]	sys_idISNOTEMPTY*EQ	false	true	2015-08-21 17:28:31	
Hide meta fields	label [sc_ic_question]		true	true	2014-01-21 12:43:35	
Set Table column ReadOnly	To-dos Widget Mapping [sn_hr_sp_todos_widget_mapping]		true	true	2018-08-02 15:16:10	
Clear save signature before action execution value - when precondition is not Signature	Function [sys_sg_button]	preconditions!=signature*EQ	true	true	2019-04-18 07:01:21	
Content vs parameters	HTTP Method [sys_rest_message_fn]	http_method=get*ORhttp_method=delete*ORh...	true	true	2016-12-14 16:50:13	



UI Policy - High Impact Control

Table: Incident [incident]

Application: Global

Active: ☒

* Short description: High Impact Control

Order: 100

When to Apply: Script

Conditions: Add Filter Condition Add OR Clause

Impact is 1 - High

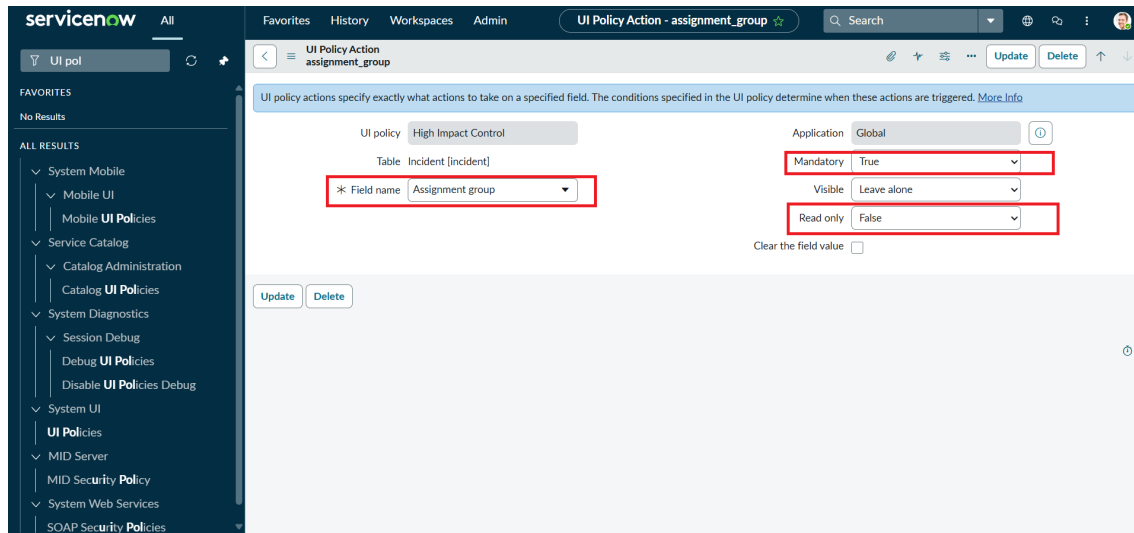
Reverse if false: ☒

On load: ☒

Inherit: ☐

Activity 2: Add UI Policy Action for Assignment group

- Open the High Impact Control UI Policy and scroll down to the UI Policy Actions related list.
- Click on New.
- Select Field name as “Assignment group”.
- Set Mandatory as “True”, Visible as “Leave alone”, and Read only as “False”.
- Click on Submit to save the action.



The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar displays the navigation menu with 'UI Policies' expanded. The main content area shows the configuration for a UI Policy Action named 'assignment_group' under the 'High Impact Control' policy. The configuration includes a table 'Incident [incident]' with a field 'Assignment group' selected. The 'Mandatory' checkbox is checked, 'Visible' is set to 'Leave alone', and 'Read only' is set to 'False'. The 'Update' and 'Delete' buttons are visible at the bottom.

UI policy: High Impact Control

Application: Global

Table: Incident [incident]

* Field name: Assignment group

Mandatory: True

Visible: Leave alone

Read only: False

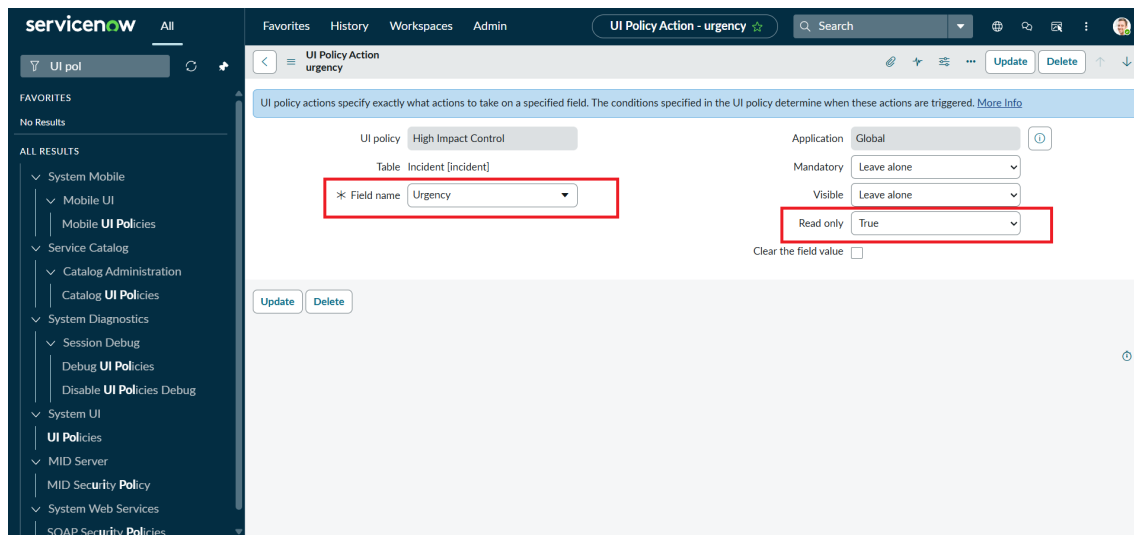
Clear the field value: ☐

Update Delete

Milestone 2 : UI Policy Action – Urgency

Activity 1: Configure the Urgency field

- Go to the High Impact Control UI Policy and scroll down to the UI Policy Actions related list.
- Click on New to create a UI Policy Action.
- Select Field name as “Urgency”.
- Set Read only as “True”, so Urgency cannot be changed whenever Impact is High.
- Leave Visible as “Leave alone” so the field stays visible on the form.
- Click on Submit to save the action.
- To verify, open an Incident and set Impact to High. Urgency becomes read-only. Changing Impact to another value reverts it.



The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar shows the navigation menu with 'UI Policies' expanded. The main content area is titled 'UI Policy Action - urgency'. The configuration details are as follows:

- UI policy:** High Impact Control
- Table:** Incident [Incident]
- * Field name:** Urgency
- Application:** Global
- Mandatory:** Leave alone
- Visible:** Leave alone
- Read only:** True
- Clear the field value:** ☐

Buttons for 'Update' and 'Delete' are visible at the bottom of the configuration area.

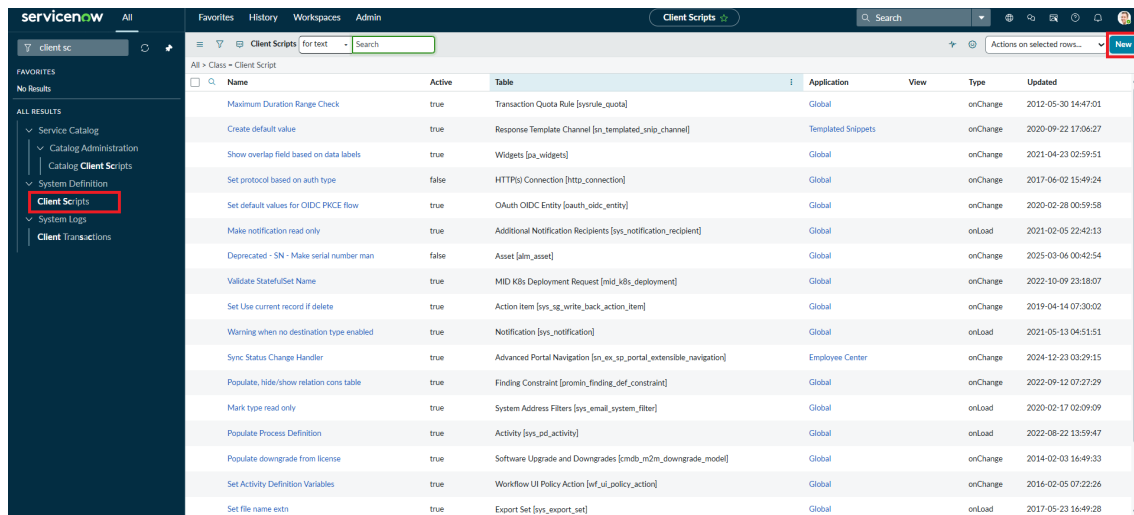
Milestone 3 : onChange Client Script

Activity 1: Create Client Script to auto set Urgency

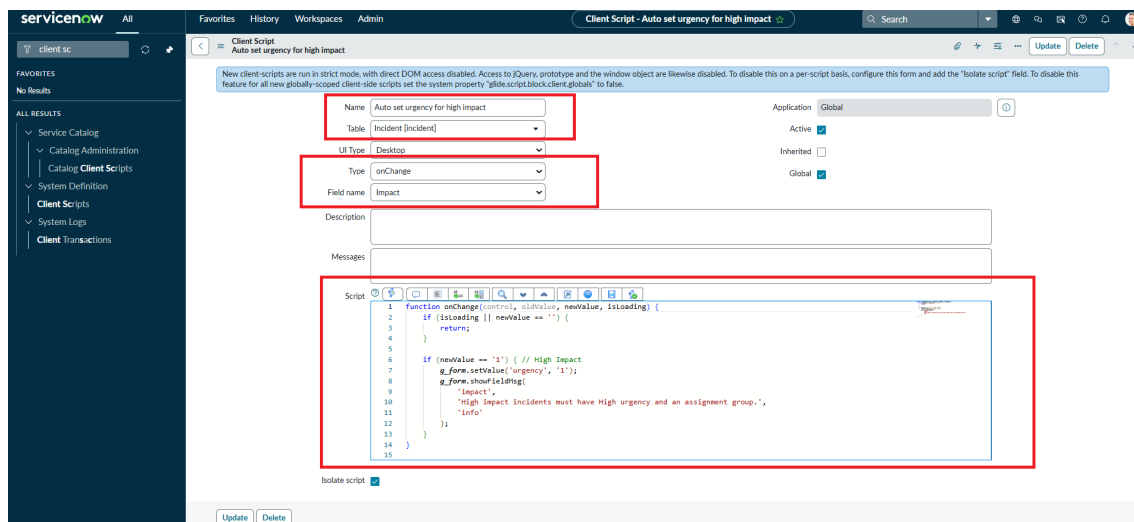
- Click on All >> search for System UI >> click on Client Scripts.
- Click on New to create a new Client Script.
- Give Name as “Auto set urgency for high impact”.
- Set Table as “Incident [incident]”, Type as “onChange”, and Field name as “Impact”.
- Keep Active as true.
- Paste the following script in the Script field and click on Submit.

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (newValue == '1') {
        g_form.setValue('urgency', '1');
        g_form.addInfoMessage('Urgency set to High for High impact incident.');
```

To verify, open an Incident and change Impact to High. The Urgency field updates automatically.



Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [rn_templated_snip_channel]	Templated Snippets		onChange	2020-09-22 17:06:27
Show overlap field based on data labels	true	Widgets [pa_widgets]	Global		onChange	2021-04-23 02:59:51
Set protocol based on auth type	false	HTTP(s) Connection [http_connection]	Global		onChange	2017-06-02 15:49:24
Set default values for OIDC PKCE flow	true	OAuth OIDC Entity [oauth_oidc_entity]	Global		onChange	2020-02-28 00:59:58
Make notification read only	true	Additional Notification Recipients [sys_notification_recipient]	Global		onLoad	2021-02-05 22:42:13
Deprecated - SN - Make serial number man	false	Asset [asn_asset]	Global		onChange	2025-03-04 00:42:54
Validate State/Set Name	true	MID Kbs Deployment Request [mid_kbs_deployment]	Global		onChange	2022-10-09 23:18:07
Set Use current record if delete	true	Action Item [sys_sg_write_back_action_item]	Global		onChange	2019-04-14 07:30:02
Warning when no destination type enabled	true	Notification [sys_notification]	Global		onLoad	2021-05-13 04:51:51
Sync Status Change Handler	true	Advanced Portal Navigation [rn_ex_sp_portal_extensible_navigation]	Employee Center		onChange	2024-12-23 03:29:15
Populate, hide/show relation cons table	true	Finding Constraint [promin_finding_def_constraint]	Global		onChange	2022-09-12 07:27:29
Mark type read only	true	System Address Filters [sys_email_system_filter]	Global		onLoad	2020-02-17 02:09:09
Populate Process Definition	true	Activity [sys_pd_activity]	Global		onLoad	2022-08-22 13:59:47
Populate downgrade from license	true	Software Upgrade and Downgrades [smdb_m2m_downgrade_model]	Global		onChange	2014-02-03 16:49:33
Set Activity Definition Variables	true	Workflow UI Policy Action [wf_ui_policy_action]	Global		onChange	2016-02-05 07:22:26
Set file name extn	true	Export Set [sys_export_set]	Global		onLoad	2017-05-23 16:49:28



Client Script - Auto set urgency for high impact

Name: Auto set urgency for high impact

Table: Incident [incident]

UI Type: Desktop

Type: onChange

Field name: Impact

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script:

```
1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || newValue == '') {
3     return;
4   }
5   if (newValue == '1') { // High Impact
6     g_form.setValue('urgency', '1');
7     g_form.showFieldMsg(
8       'Impact',
9       'High impact incidents must have High urgency and an assignment group.',
10      'Info'
11    );
12  }
13 }
14
15
```

Isolate script: ☒

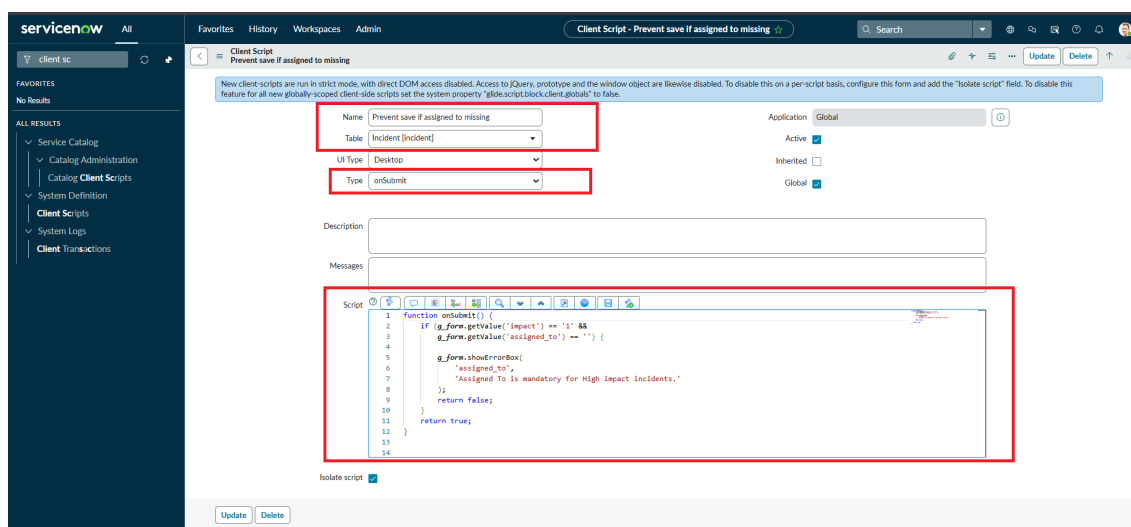
Milestone 4 : onSubmit Client Script (Save Validation)

Activity 1: Create Client Script to prevent save if Assigned to is missing

- Click on All >> System UI >> Client Scripts, then click on New.
- Give Name as “Prevent save if assigned to missing”.
- Set Table as “Incident [incident]” and Type as “onSubmit”.
- Keep Active as true.
- Paste the following script in the Script field and click on Submit.

```
function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
        g_form.getValue('assigned_to') == '') {
        g_form.showErrorBox(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
        return false;
    }
    return true;
}
```

To verify, try saving an Incident with High impact and no Assigned to. The save is blocked with an error message.



Milestone 5 : onCellEdit Client Script

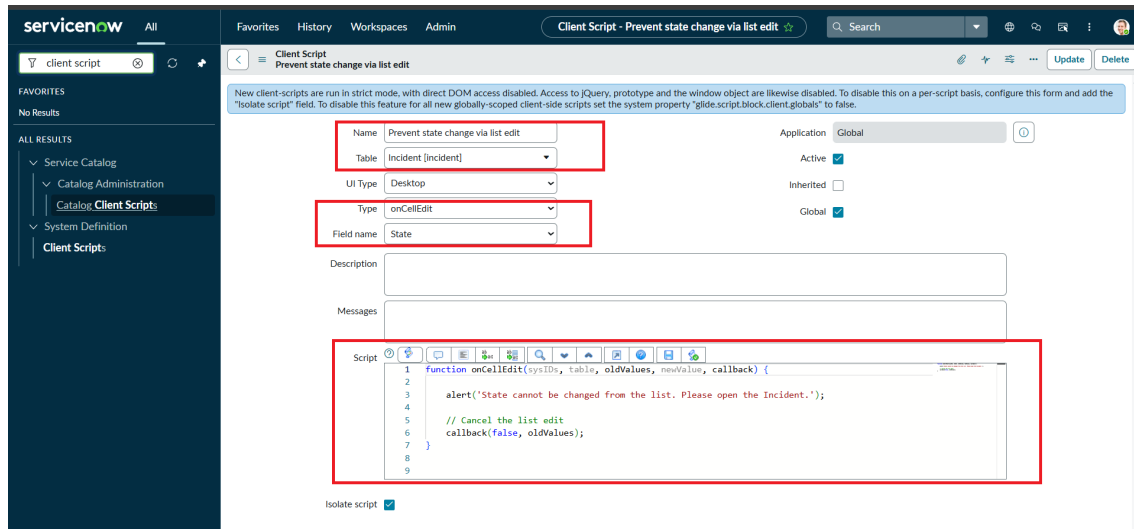
Activity 1: Create Client Script to prevent State change from list

- Click on All >> System UI >> Client Scripts, then click on New.
- Give Name as “Prevent state change via list edit”.
- Set Table as “Incident [incident]”, Type as “onCellEdit”, and Field name as “State”.
- Keep Active as true.
- Paste the following script in the Script field and click on Submit.

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
    alert('State cannot be updated using list editing. Please open the Incident.');
```

```
    callback(false);
}
```

To verify, open the Incident list and try to change the State field directly. The change is blocked.



The screenshot shows the ServiceNow interface for configuring a Client Script. The script is named 'Prevent state change via list edit' and is associated with the 'Incident' table. The script type is 'onCellEdit' and the field name is 'State'. The script is active and global. The script content is as follows:

```
1 function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
2
3     alert('State cannot be changed from the list. Please open the Incident.');
```

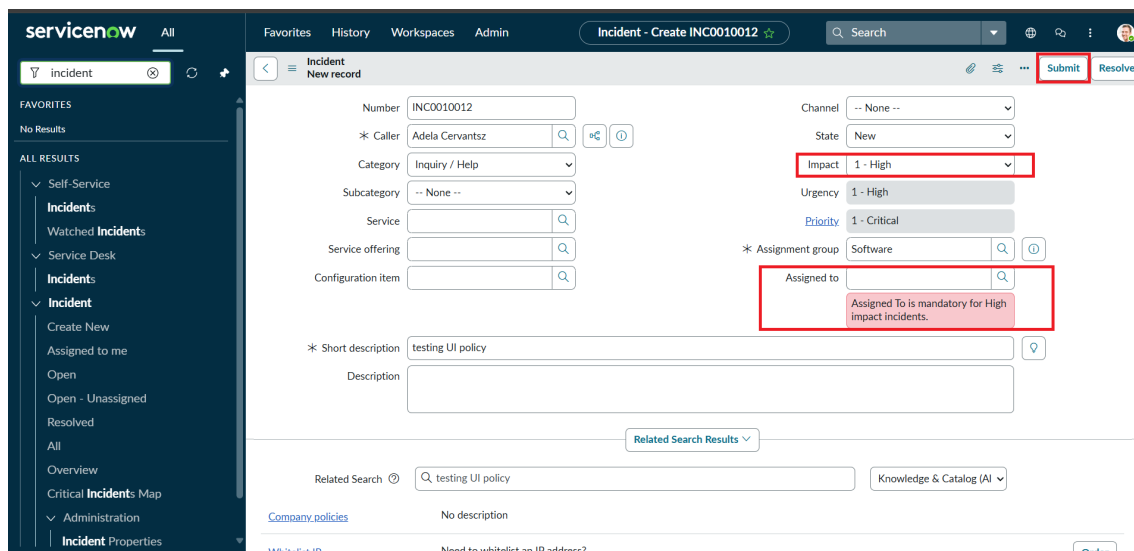
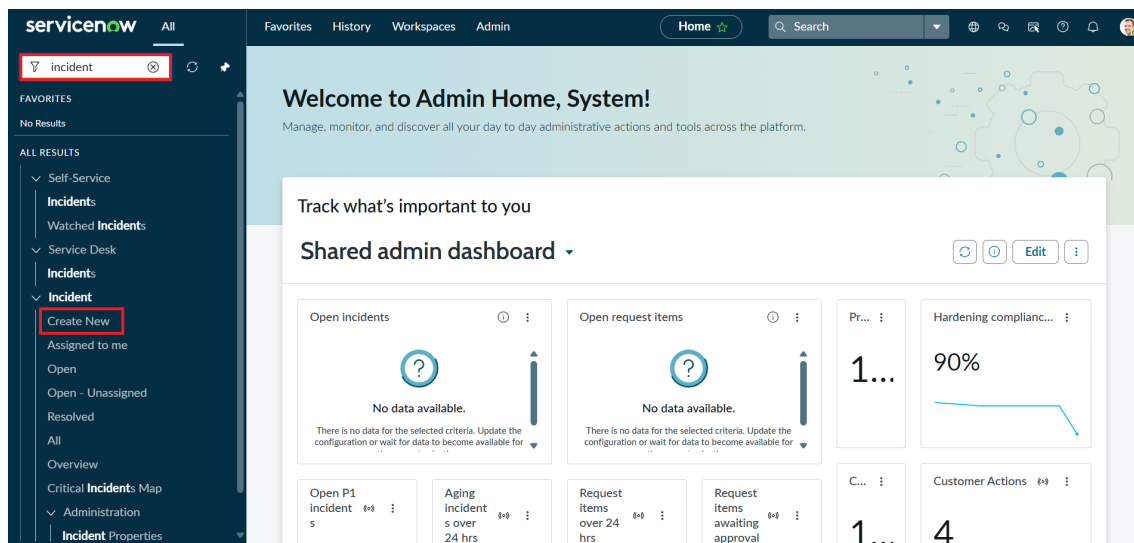
```
4
5     // Cancel the list edit
6     callback(false, oldValues);
7 }
8
9
```

The 'Isolate script' checkbox is checked.

Milestone 6 : Testing the Configuration

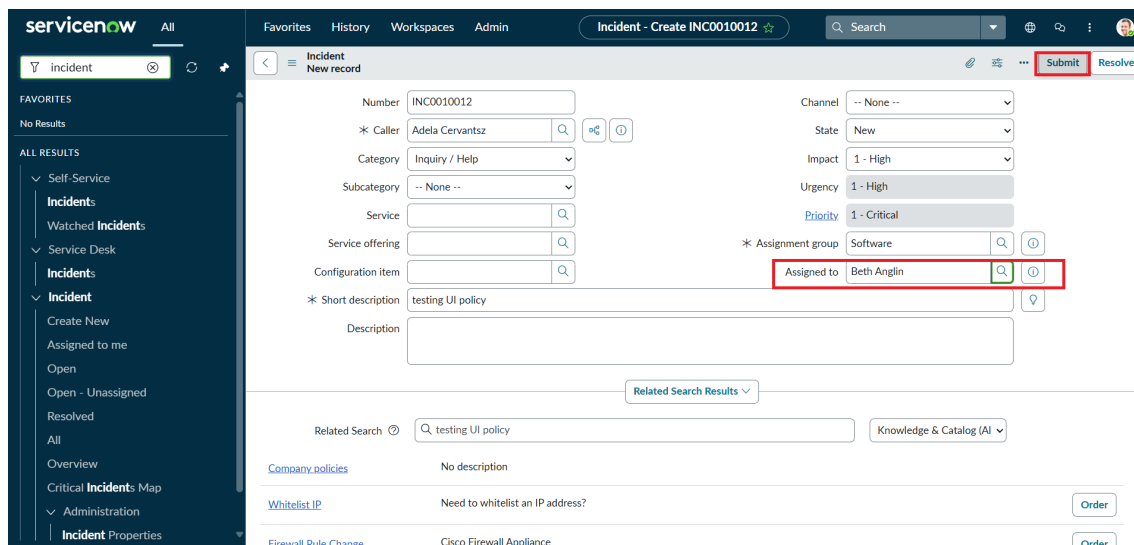
Activity 1: Test Mandatory Enforcement

- Go to Incident >> Create New to open a new Incident form.
- Select “1 - High” for the Impact field. Urgency is set to High and becomes read-only.
- Leave the Assigned to field empty.
- Click on Submit.
- The Incident is not saved, and an error message shows that Assigned to is mandatory for High impact incidents.



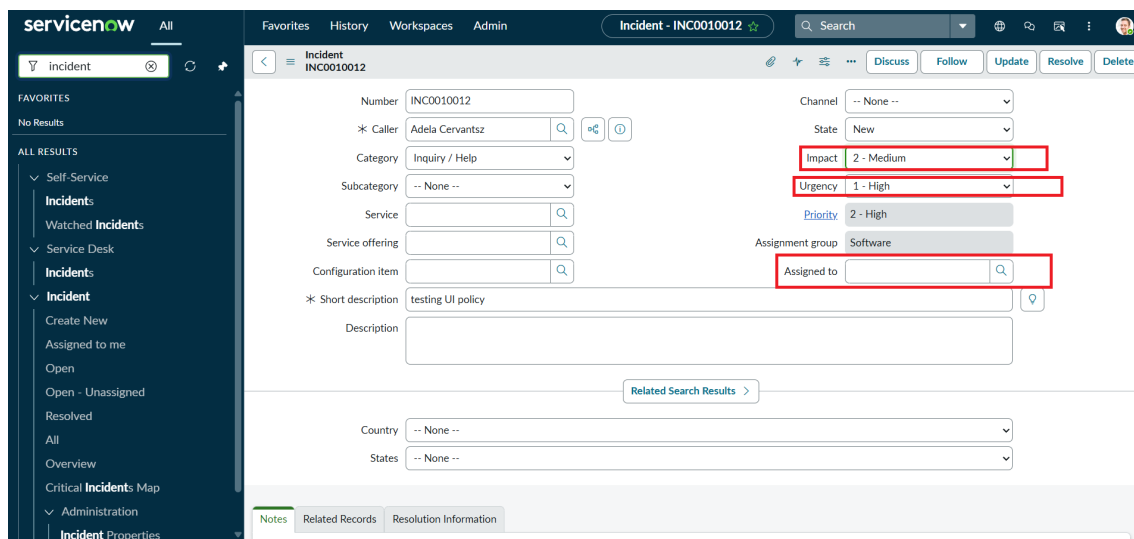
Activity 2: Test Successful Save

- Open the Incident form and select a user in the Assigned to field.
- Click on Submit.
- The Incident is saved successfully without any errors.



Activity 3: Reverse Condition Test

- Go to Incident >> Open and select an Incident where Impact is High.
- Change Impact from High to Medium.
- Assigned to is no longer mandatory, and Urgency is editable again as the Reverse if false option takes effect.
- Click on Update. The record saves without errors.



Activity 4: Test List Edit Blocking

- Go to Incident >> All to view the Incident list.
- Double-click the State field of any Incident to edit it directly in the list.
- An alert appears saying State cannot be changed from the list, and the State value remains unchanged.

servicenow All

Favorites History Workspaces Admin Incidents

Incident

Incidents

Number

Search

Actions on selected rows... New

All > Active = true

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	David Miller	3 -					2016-08-10	admin

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Favorites History Workspaces Admin Incidents

Incident

Incidents

Number

Search

Actions on selected rows... New

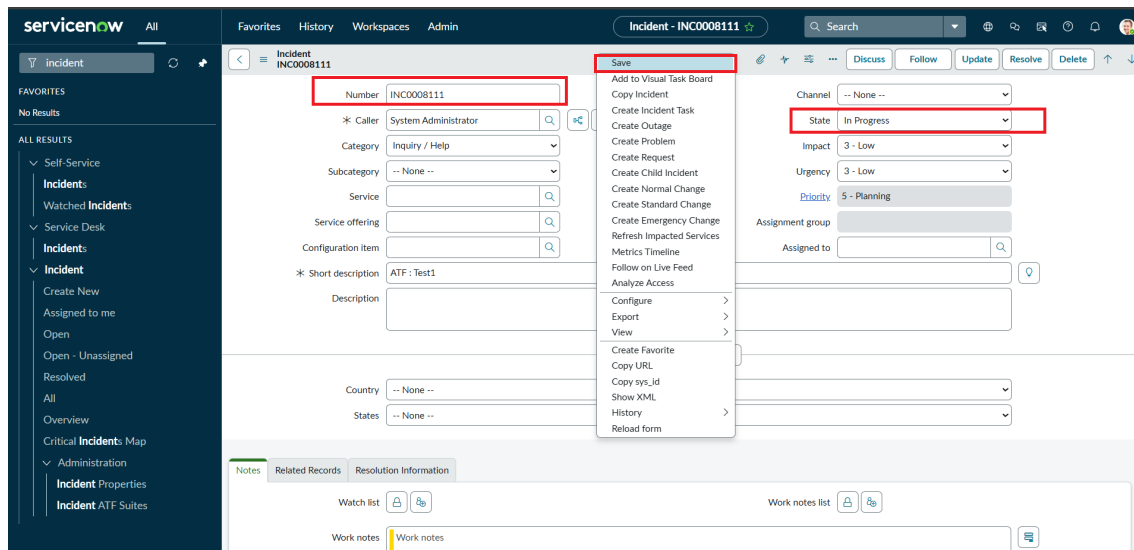
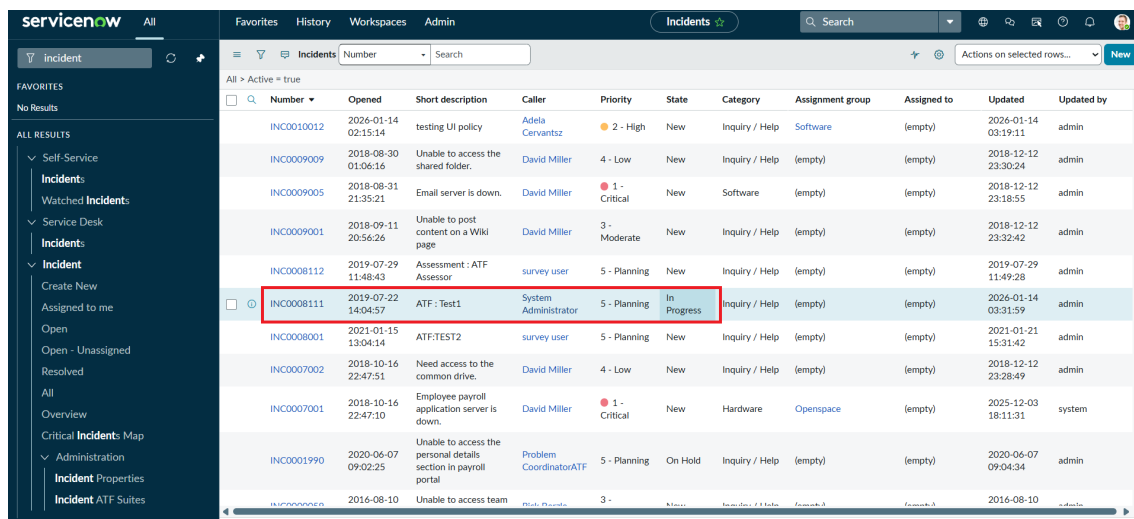
All > Active = true

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	David Miller	3 -					2016-08-10	admin

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Activity 5: Test Form-Based Update

- Open the same Incident record and change the State on the Incident form.
- Click on Update.
- The State change is saved successfully, and other fields still follow the UI Policy and Client Script rules.

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF - Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-01-14 03:31:59	admin
INC0008001	2021-01-15 13:04:14	ATF-TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	Private Practice	3 -	Minor	Research & Analysis	Research & Analysis	(empty)	2016-08-10	admin

Conclusion :

The Implement Client Script & UI Policy (Incident) project demonstrates how UI Policies and Client Scripts work together to enforce dynamic field behaviour, automate updates, and prevent incorrect data submission on Incident forms. By using onChange, onSubmit, and onCellEdit scripts alongside UI Policy actions, fields such as Assigned to and Urgency respond appropriately to changes in Impact, keeping data clean and consistent. The solution enforces mandatory fields and controls user interactions while remaining lightweight and easy to implement, and it shows practical ways to improve form usability and data integrity in ServiceNow.